COSC264
Introduction to Computer Networks and the Internet

Application Protocols: Introduction to the Web and HTTP

Dr Barry Wu

WRC
University of Canterbury
Barry.wu@canterbury.ac.nz

Learning Objectives

- At the end of the lectures, you will be able to:
  - Understand the concept of network applications
  - Explain the role of HTTP in the operation of the Web
  - Describe the HTTP message format
  - Explain the difference between non-persistent and persistent connections
  - Explain cookies
  - Explain caching

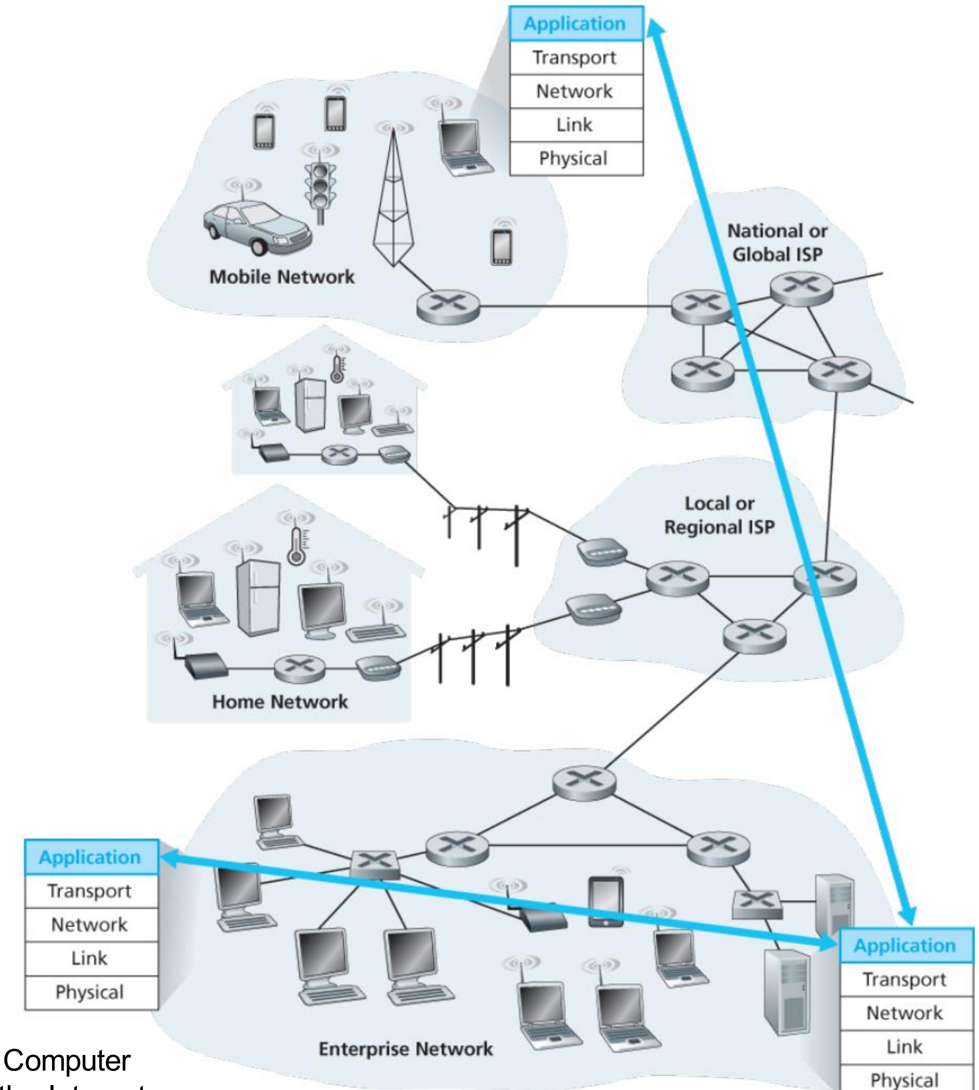
Outline

- Network applications
- The Web
- HTTP
  - Non-persistent and persistent connections
  - Messages
  - Cookies
  - Web caching

Network Applications

What is a network application?

- Programs that run on different end systems and communicate with each other over the network



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.1.

Some Popular Network Applications

Web browsing

- Google Chrome, Mozilla Firefox, Microsoft Edge

Email

- Gmail, Outlook

Remote access

- PuTTY, Remote Desktop

Instant messaging

- Slack, Microsoft Teams

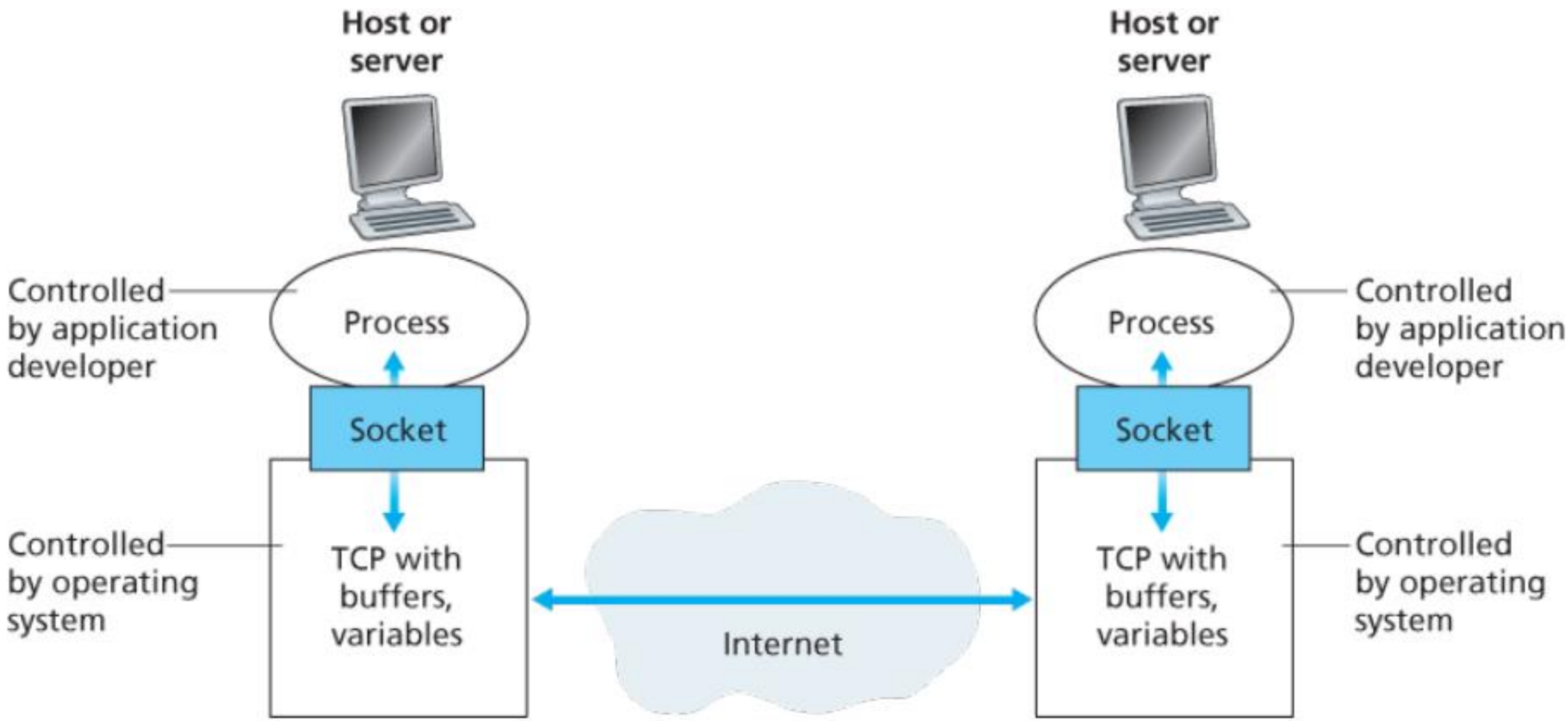
Streaming services

- Netflix, Spotify, YouTube

File transfer

- FileZilla, WinSCP

Application processes, sockets, and underlying transport protocol



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.3.

Application-Layer Protocols

- An application-layer protocol defines how an application's processes, running on different end systems, pass messages to each other.
- An application-layer protocol is only one piece (a big piece) of a network application.

Network application	Application protocol
Web application	HTTP
Email	SMTP

Services an Application Needs from a Transport-Layer Protocol

- <u>Reliable data transfer</u>

Applications that require reliable data transfer	• Email, instant messaging, file transfer, financial applications, etc.
Applications that are loss-tolerant	• Multimedia applications, etc.

Services an Application Needs from a Transport-Layer Protocol (cont.)

- Bandwidth

Bandwidth-sensitive applications	• Internet telephony applications, multimedia applications, etc.
Elastic applications	• Email, file transfer, etc.

Services an Application Needs from a Transport-Layer Protocol (cont.)

- Timing
  - End-to-end delays

Time-sensitive applications	• Interactive real-time applications, such as Internet telephony, virtual environments, teleconferencing, multiplayer games, etc.
Non time-sensitive applications	• Email, file transfer, etc.

Services Provided by the Internet Transport Protocols

TCP	UDP
Connection-oriented service	Connectionless service
Reliable transport service	*Unreliable data transfer service*
Congestion control	*No congestion control*
Flow control	*No flow control*
No guarantee on the rate of delivery *No guarantee on the delays*	

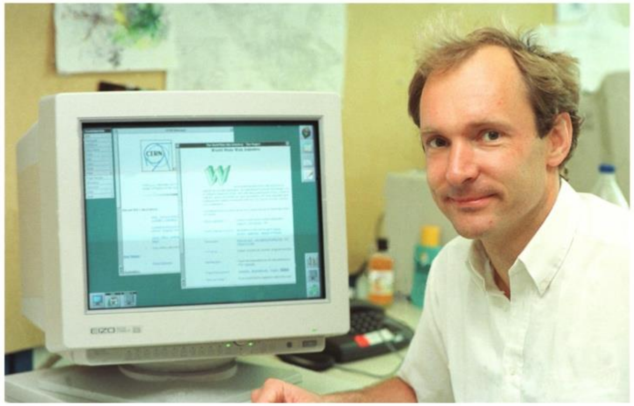
The Internet has been hosting time-sensitive applications for many years!

The Web and HTTP

World Wide Web

- The World Wide Web, commonly known as the Web, is a global collection of documents and other resources to be shared over the Internet.

- Tim Berners-Lee, a British scientist, invented the World Wide Web (WWW) in 1989.



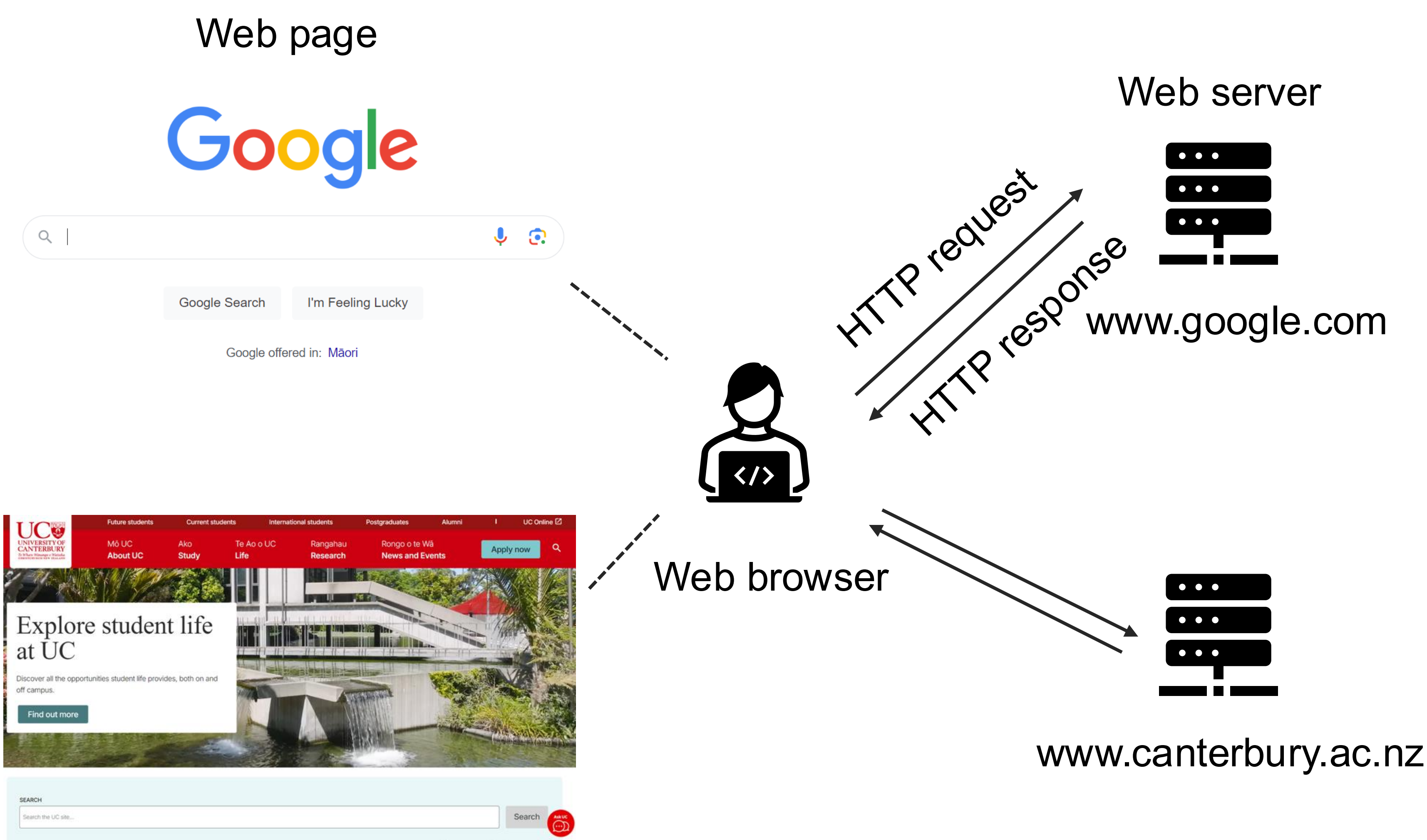
Tim Berners-Lee, pictured at CERN (Image: CERN)

A replica of the NeXT machine used by Tim Berners-Lee in 1990 to develop and run the first WWW server, multimedia browser and web editor (Image: Maximilien Brice/Anna Pantelia/CERN)

Source: https://home.cern/science/computing/birth-web/short-history-web

World Wide Web (cont.)

- Architecture of the Web



HTTP Overview

- The Hypertext Transfer Protocol (HTTP) is the foundation protocol of the Web

Protocol	Published	Status	RFC
HTTP/1.0	1996	Obsolete	RFC 1945
HTTP/1.1	1997	Standard	RFC 2068, 2616, 7230-7235
HTTP/2	2015	Standard	RFC 7540
HTTP/3	2022	Standard	RFC 9114

Source: https://en.wikipedia.org/wiki/HTTP

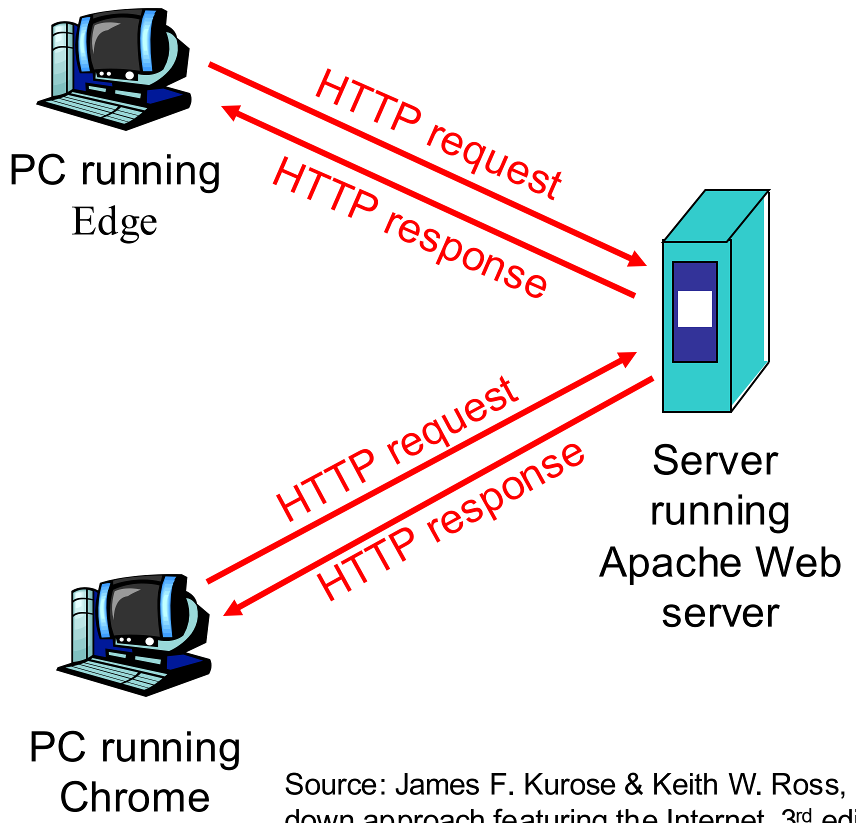
HTTP Overview (cont.)

- A Web page consists of objects
- An object is a file (e.g., an HTML file, a JPEG image)
- Most Web pages consist of a base HTML file and several referenced objects
- Each object is addressable by a URL

$$\texttt{http://}\underbrace{\texttt{www.someschool.edu}}_{\text{host name}}\underbrace{\texttt{/someDept/pic.gif}}_{\text{path name}}$$

HTTP Overview (cont.)

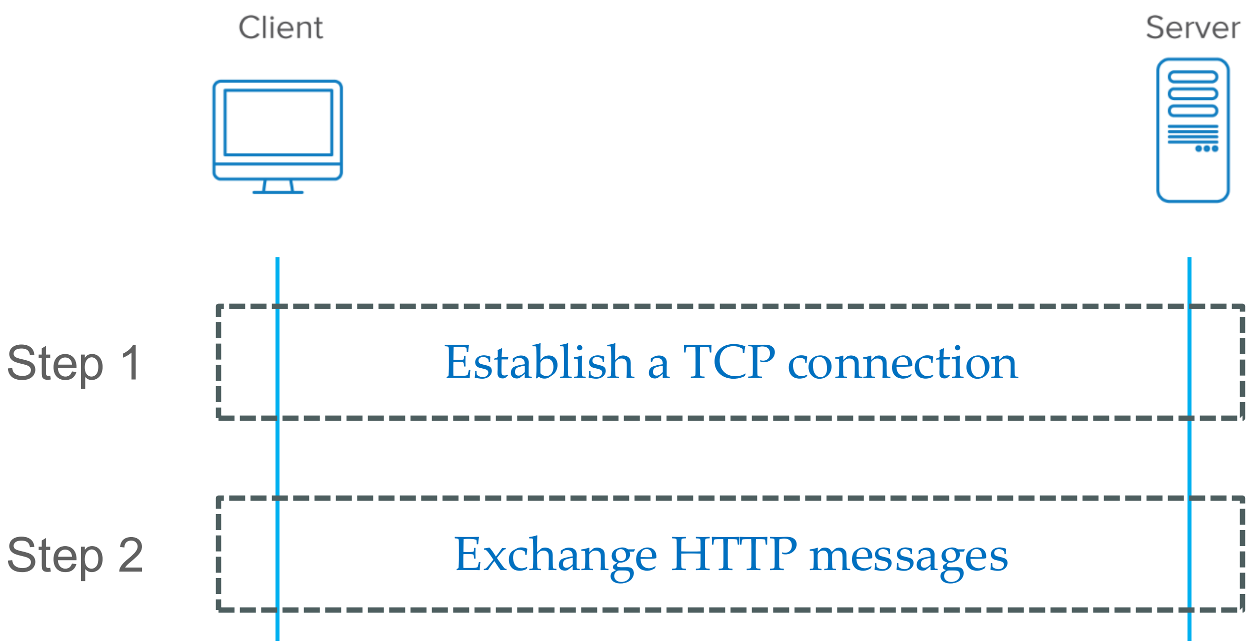
- Client/server model
  - *Client:* Web browser that requests, receives, and displays Web objects
  - *Server:* Web server that sends objects in response to requests



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.6.

HTTP Overview (cont.)

- HTTP is stateless
  - An HTTP server can work without maintaining any information about past client requests.
- HTTP uses TCP (*HTTP/3 is an exception*)



HTTP Connections

Non-persistent HTTP

- At most **one object** is sent over a TCP connection.
- HTTP/1.0 uses nonpersistent HTTP.

Persistent HTTP

- **Multiple objects** can be sent over a TCP connection.
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP.

HTTP with Non-Persistent Connections

Suppose user enters the URL

`www.someSchool.edu/someDepartment/home.index`

The HTTP client initiates a TCP connection to the server www.someSchool.edu on port number 80

↓

The HTTP client sends an HTTP request message to the server via its socket. The request message includes the path name /someDepartment/home.index

↓

The HTTP server receives the request message via its socket, retrieves the object from its storage, encapsulates the object in an HTTP response message, and sends the response to the client via its socket

↓

The HTTP server tells TCP to close the TCP connection

↓

The HTTP client receives the response message. The TCP connection terminates. The client extracts the file from the response message, examines the HTML file, and finds *references* to the 10 JPEG objects.

Repeat for each referenced JPEG object

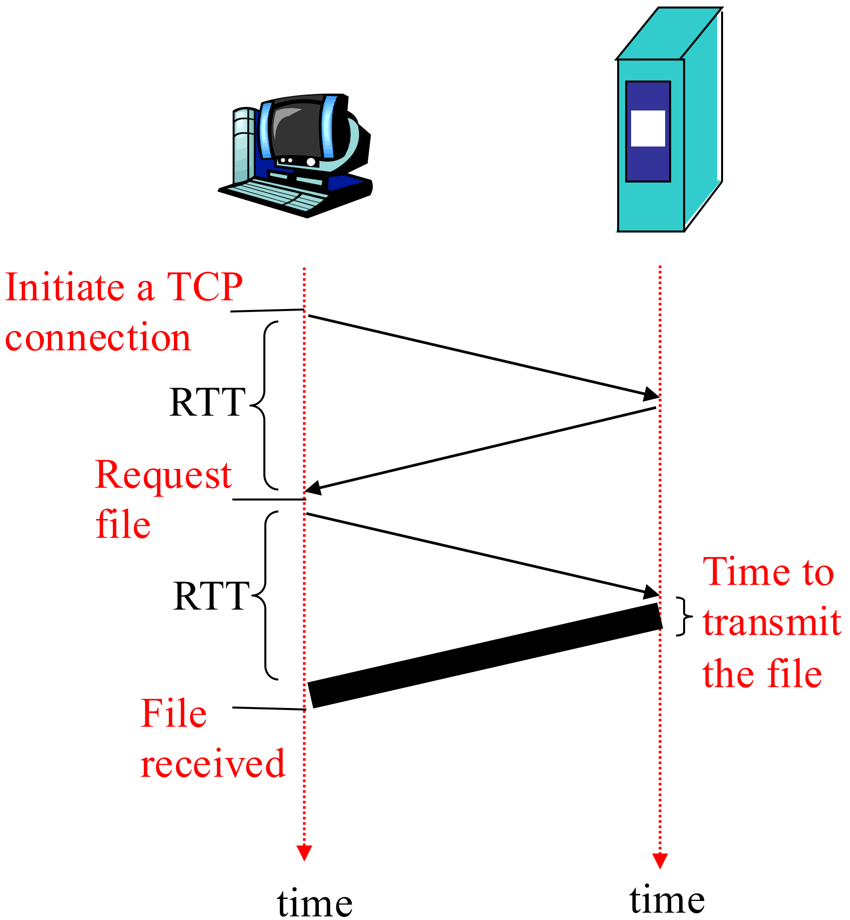
Response Time Modeling

Definition of RTT (Round Trip Time)

- time to send a small packet to travel from the client to the server and back to the client

Response time

- one RTT to initiate a TCP connection
- one RTT for HTTP request and response
- file transmission time

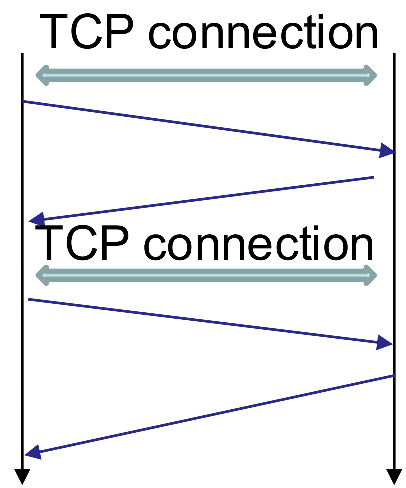


Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.7.

Persistent HTTP

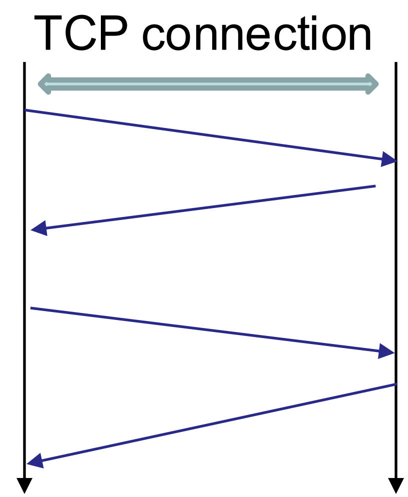
Non-persistent HTTP

- require 2 RTTs per object
- resources are allocated for each TCP connection – a burden for the server especially.



Persistent HTTP

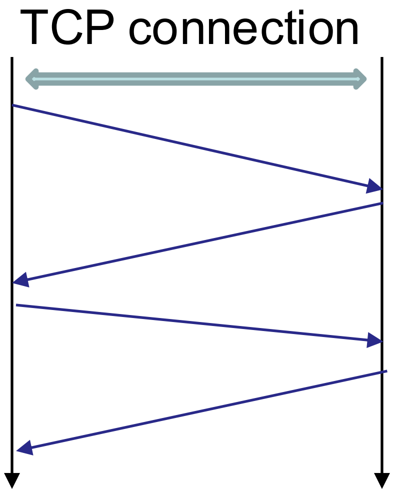
- the server leaves connection open after sending a response
- subsequent HTTP messages between the same client and server are sent over the same connection



Persistent HTTP (cont.)

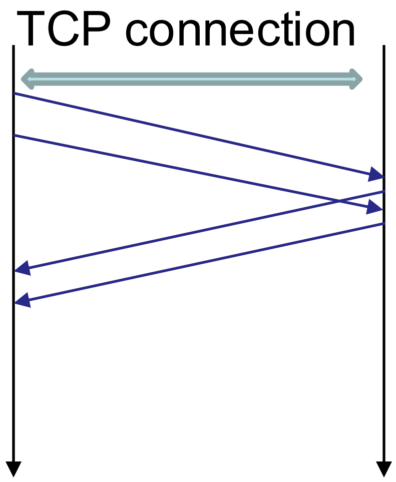
Persistent without pipelining

- the client issues new request only when previous response has been received
- one RTT for each referenced object



Persistent with pipelining

- default in HTTP/1.1
- the client sends requests as soon as it encounters a referenced object



HTTP request message

- HTTP request message:

URL field

HTTP version field

request line (GET, POST, HEAD commands)

```
GET /somedir/page.html HTTP/1.1
Host: www.someschool.edu
User-agent: Mozilla/4.0
Connection: close
Accept-language:fr
```

header lines

Nonpersistent connections

Carriage return, blank line

(Entity body)

HTTP Request Message - Accept-Language Header

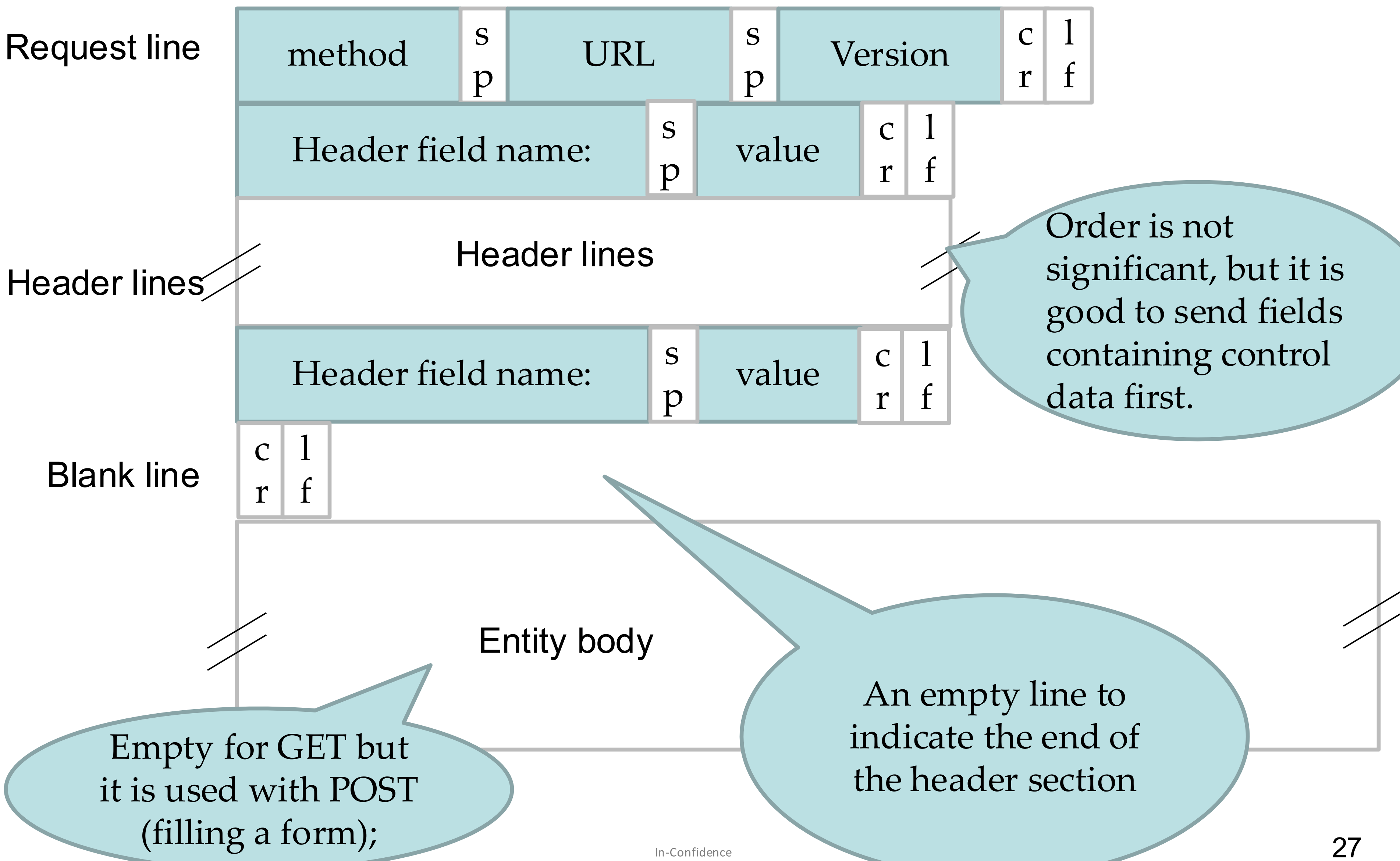
Accept-Language: <language-range>[;q=<weight>], <language-range>[;q=<weight>], ...

Each language-range MAY be given an associated quality value which represents an estimate of the user's preference for the languages specified by that range. The quality value defaults to "q=1".

For example,
Accept-Language: da, en-gb;q=0.8, en;q=0.7

would mean: "*I prefer Danish, but will accept British English and other types of English*."

General format of a request message



HTTP Request Message – Request Methods

- Some built-in HTTP request methods

Method	Description
GET	Read a Web page
HEAD	Read a Web page's header
POST	Append to a Web page
PUT	Store a Web page
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Connect through a proxy
OPTIONS	Query options for a page

Source: Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, 5th edition, 2010, Figure 7-37.

HTTP response message

status line (protocol ver, status code, status msg.)

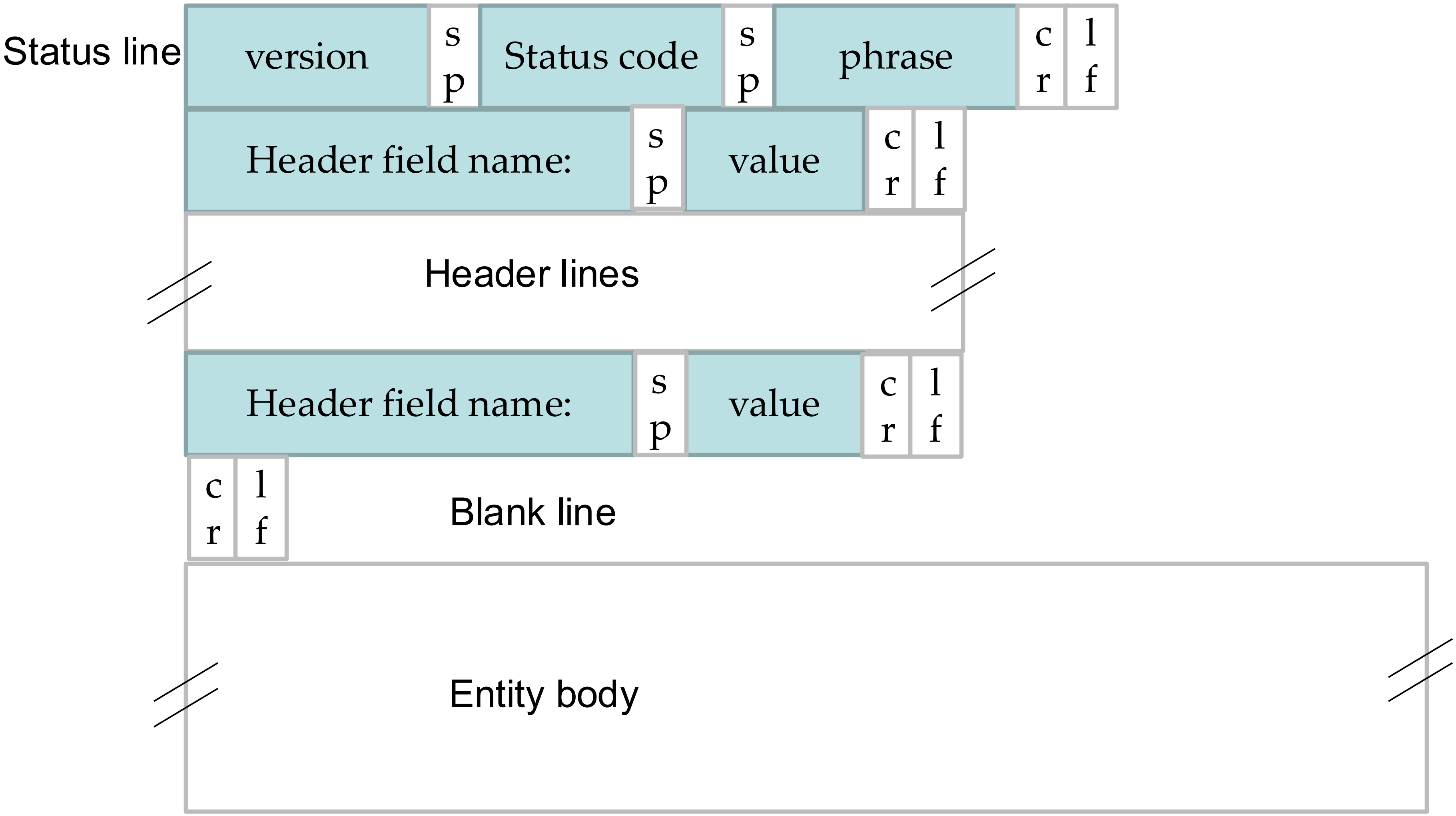
header lines

data, e.g., requested HTML file

```
HTTP/1.1 200 OK
Connection close
Date: Thu, 06 Aug 1998 12:00:15 GMT
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 1998 …...
Content-Length: 6821
Content-Type: text/html

data data data data data ...
```

General format of a response message



In-Confidence

HTTP Response Status Codes

200 OK

- request succeeded and requested object is returned in the response

301 Moved Permanently

- requested object has been permanently moved; the new location is specified in the response message (Location:)

400 Bad Request

- request not understood by the server

404 Not Found

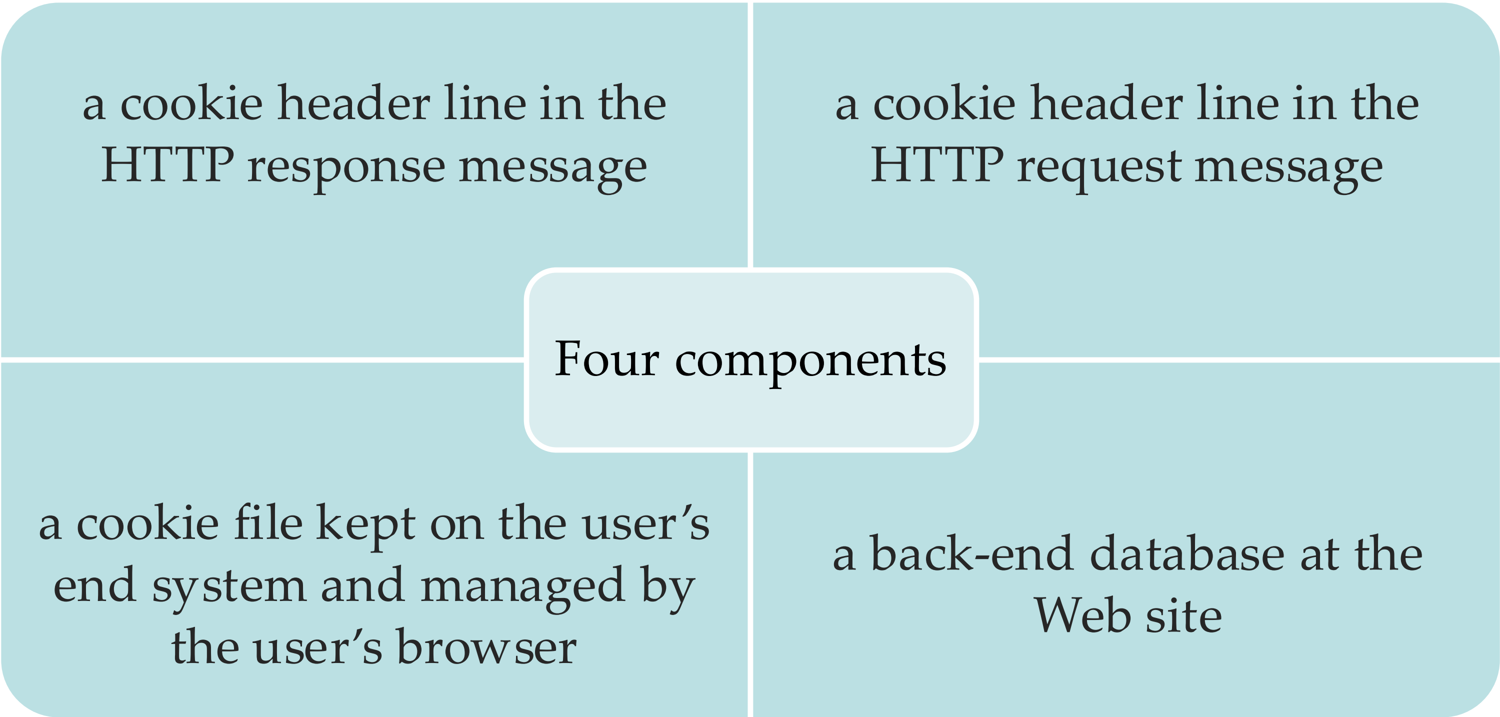
- requested document not found on the server

505 HTTP Version Not Supported

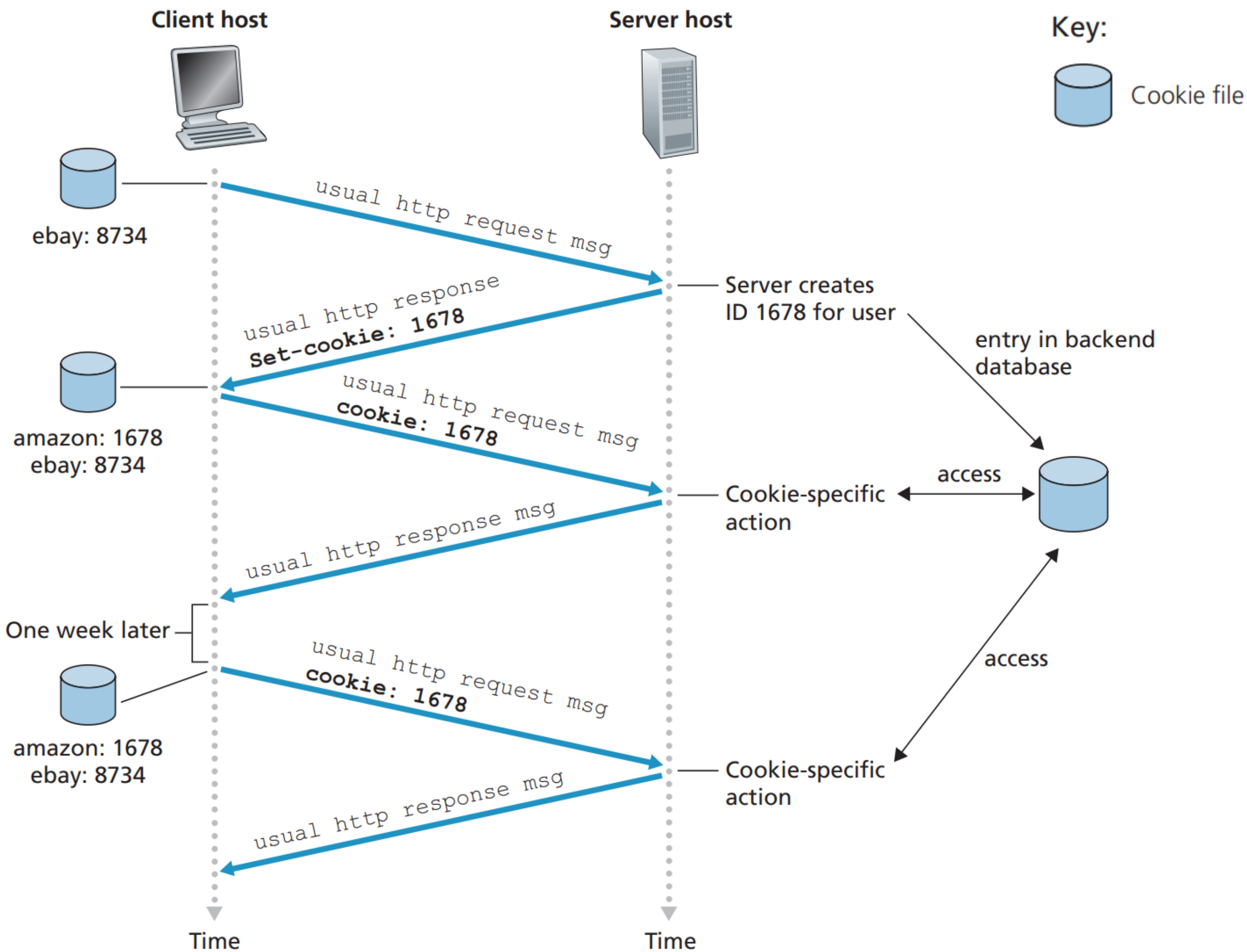
- requested HTTP protocol version is not supported by the server

User-Server Interaction: Cookies [RFC 2109]

- Many major Web sites use cookies to identify users
  -  Online shopping



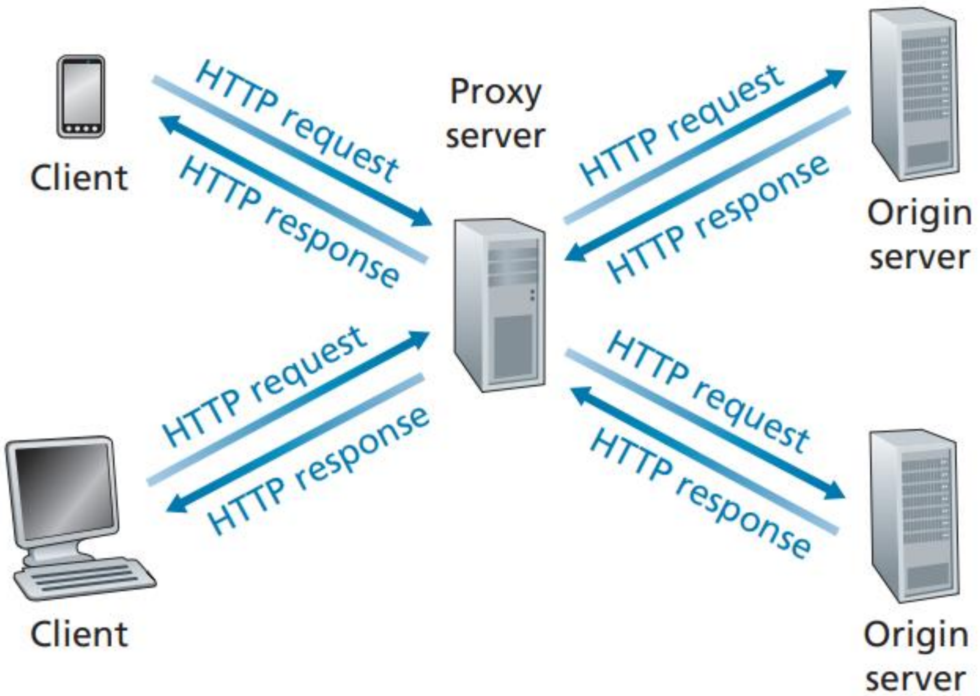
User-Server Interaction: Cookies (cont.)



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, , 8th edition, 2021, Figure 2.10

Caching

- A Web cache, also called a proxy server, is a network entity that satisfies HTTP requests on behalf of an origin Web server.



A cache is both a server and a client at the same time.

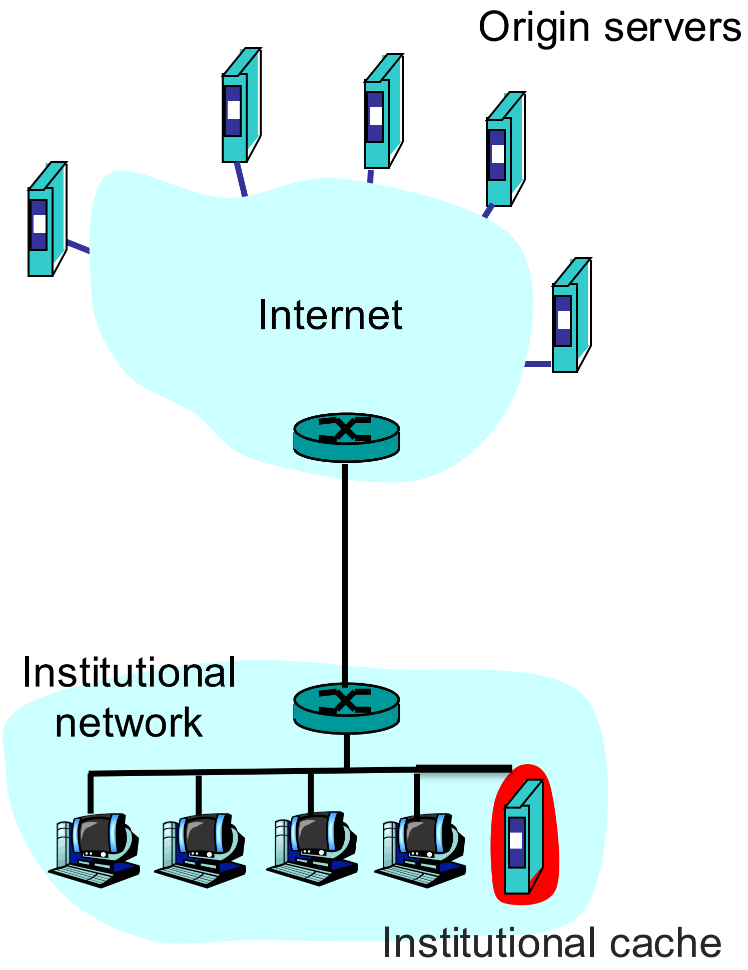
- When it receives requests from and sends responses to a browser, it is a server.
- When it sends requests to and receives responses from an origin server, it is a client.

Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.10.

Caching (cont.)

Why caching?

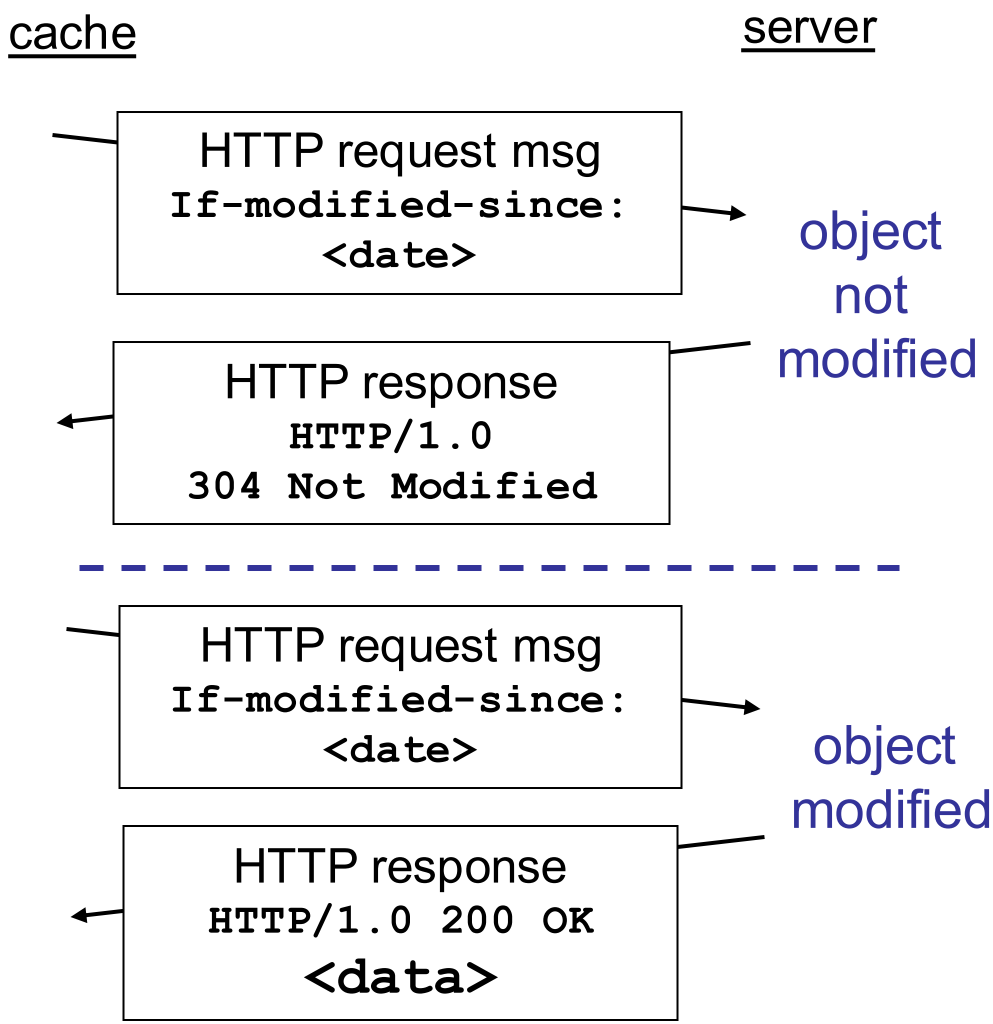
- Reduce the response time for a client request
- Reduce traffic on an institution's access link to the Internet
- Lower bandwidth costs



Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 2.12.

Conditional GET

- Goal: don't send object if cache has up-to-date cached version
- cache: specify date of cached copy in HTTP request
  `If-modified-since: <date>`
- server: response contains no object if cached copy is up-to-date:
  `HTTP/1.0 304 Not Modified`



References

- James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, Addison Wesley, 3rd edition, 2005.
  - Chapter 2 Application Layer (2.1, 2.2)
- William Stallings, Data and Computer Communications, Pearson, 10th edition, 2013.
  - Chapter 24 Electronic Mail, DNS, and HTTP (24.3)
- Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, Prentice Hall, 5th edition, 2010.
  - Chapter 7 The Application Layer (7.3)

Acknowledgements

- Lecture notes are developed based on slides from the following three sources:
  - Dr Mengmeng Ge’s lecture notes for COSC264, University of Canterbury, 2023
  - Assoc Prof Dan Kim’s lecture notes for COSC264, University of Canterbury, 2017
  - Prof Aleksandar Kuzmanovic’s lecture notes for CS340, Northwestern University, 2005, https://users.cs.northwestern.edu/~akuzma/classes/CS340-w05/lecture_notes.htm